SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5101

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 **Total Marks:50**

Annexure A

CONCEPT MAPPING

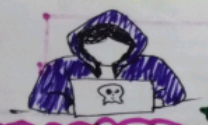
Code of Conduct:

I Bharath. M (192324302) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Bharath

25/100

CONCEPT MAP - Security Attacks, Theats, Risks And Security Goals



SECURITY ATTACKS
(Actions by an attacker to gain unauthorized access & manipulate data)

Eavesdropping
- Intercepting data during Transmission
- Example: Packet Sniffing on public Wi-Fi

Tampering
- Modifying without permission
- Example: changing a file in transit

Masquerading
- Impersonating a legitimate user/system
- Example: Phishing attack to steal credentials

Denial of Service (DOS)
- Disrupting Service availability
- Example: DDOS attack on a website

THEATS
(Potential cause of harm)

Loss of Confidentiality
(data exposure)

Data (orruption)
(Unauthorized change)

Unauthorized Access
(fake identity)

Service Disruption
(Unavailable System)

Risks
(Finacial / Operational / reputational loss)

Loss of Sensitive Information
(e.g., Personal data, credentials)

Finanical Loss
(e.g., Fraud, Fines..)

Fraud /unauthorized Transaction
(e.g. Fake Payments)

Downtime
(e.g., Loss of revenue, loss of service)

Security Goals (Midigation)

Confidentrality
Prevent unauthorized access to data

Using HTTPS to protect online banking data

Integrity
Ensure data is not altered unauthorizedly

Ex: Digital Signature ensure a PDF document is not Tampered with.

availability
Ensure System and data are accessible

Ex: Cloud backup keeps the service available even if one fails

Summary: Attacks exploit vulnerabilities → create threats → lead to risk → Security goals

SUBSTITUTION TECHNIQUES

Substitution Techniques

Caesar Cipher

A → D (k=3)

Key usage
- Single key (Shift value)
- Example : K = 3

Complexity :
- Very low
- Simple implementation

Security Strength
- Weak
- Easily broken by frequency analysis

Example :
plain : HELLO
key : 3
cipher : KHOOR

Monoalphabetic Cipher

A → Q
B → W
C → E

Key usage
- Single key (permutation of alphabet)
- 26! possible keys

Complexity :
- Low to Moderate
- Requires key Table

Security Strength
- Moderate
- Vulnerable to frequency analysis

Example
plain : HELLO
key : random mapping
cipher : QZJXC

Vigenere

A	B	C	D	E	F	
B	C	D	E	F	G	
C	D	E	F	G	H	

Key usage
- Multi-letter key
- Repeats key over plaintext

Complexity :
- Moderate
- Requires key Management

Security Strength
- Stronger than monoalphabetic
- Vulnerable if is short

Example :
plain : HELLO
key : KEY
Cipher : RIJVV

Hill Cipher

$$\begin{bmatrix} 3 & 3 & 2 \\ 2 & 5 & 7 \\ 1 & 7 & 2 \end{bmatrix}$$

Key usage
- Matrix Key (n x n)
- Uses modular arithmetic

Complexity :
- High
- Matrix Multiplion and Modular inverse

Security Strength
- Stronger
- Resists Simple frequency analysis

Example :
plain : HELLO
key : 2x2 matrix
Cipher : ZKGVQ

Most Secure Technique : Hill cipher

- Uses a matrix key and modular arithmetic, making it much harder to break.
- Provides better resistance to Simple Frequency analysis and pattern attacks.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25 20	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30 30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20 20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15 15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10 10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

